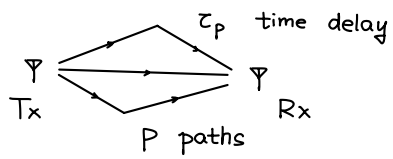
$$\xrightarrow{001101} \boxed{\text{Channel Coding}} \xrightarrow{00110101} \boxed{\text{QPSK}} \xrightarrow{(1+j),\,(-1-j),\,\cdots} \boxed{S/P} \rightarrow \begin{matrix} 1+j \\ -1-j \\ -1+j \\ \vdots \end{matrix}$$

$$\xrightarrow{\hat{x}} \boxed{\text{IFFT}} \xrightarrow{\tilde{x}} \boxed{\text{CP}} \rightarrow \boxed{P/S} \rightarrow \boxed{\text{PA}} \rightarrow \boxed{\text{上变频}} \rightarrow \boxed{W_{RF}} \xrightarrow{s} \boxed{\text{Channel}}$$

Baseband | RF Chain

Multipath : Time Delay / Selective Fading



Moving Target : Doppler / Fast Fading

OFDM 只处理 Time Delay

$$\xrightarrow{\hat{x}} \boxed{\text{IFFT}} \xrightarrow{\tilde{x}} \boxed{\text{CP}} \xrightarrow{s} \boxed{*\,h} \xrightarrow{r} \boxed{\times\,\text{CP}} \xrightarrow{\tilde{y}} \boxed{\text{FFT}} \xrightarrow{\hat{y}}$$

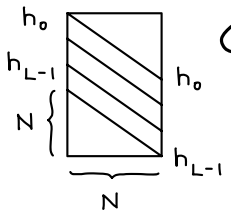
H : Toeplitz 卷积阵

$\tilde{H}$: 循环阵

$\hat{H}$: 对角阵, $\mathrm{diag}(Fh)$

$$r[m] = h * s + n = \sum_{l=0}^{L-1} h_l\, s[m-l] + n[m]$$

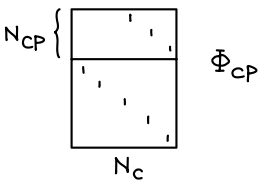
$$r_{(N+L-1)} = H_{(N+L-1)\times N}\, s_N + n$$

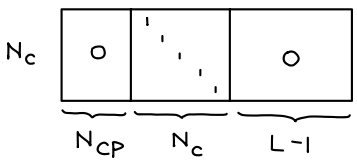


Convolution Matrix H

$h_{L\times 1}$ is sparse $\|h\|_0 = P$

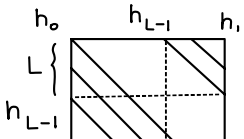
$$s_N = \underbrace{\Phi_{CP}}_{(N_c+N_{CP})\times N_c} \cdot F_{N_c}^H\, \hat{x}_{N_c} \qquad \hat{y}_{N_c} = F_{N_c} \cdot \underbrace{\Psi_{CP}}_{N_c\times(N+L-1)} \cdot r_{(N+L-1)}$$



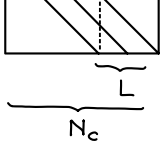


Circular $\tilde{H}$ 循环阵

$$\tilde{y} = \underbrace{\Psi_{CP}\, H\, \Phi_{CP}}_{\tilde{H}_{N_c\times N_c}}\, \tilde{x}_{N_c}$$



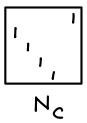
P_{th} channel delay



$$\tilde{H} = \sum_{l=1}^{L} h_l \Pi_{N_c}^{l} = \sum_{p=1}^{P} h_p \Pi_{N_c}^{\ell_p}$$

只要 $N_{CP} \geq L-1$

$\tilde{H}$ 就这样，与 N_{CP} 值无关

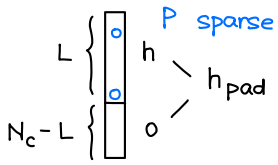


Π sparse

$$\Pi = F^H \underbrace{\begin{bmatrix} 1 & & & \\ & e^{-j2\pi/N_c} & & \\ & & \ddots & \\ & & & e^{-j2\pi(N_c-1)/N_c} \end{bmatrix}}_{\Omega} F$$

$$\hat{y} = F \Psi_{CP} (H \Phi_{CP} F^H \hat{x} + n)$$

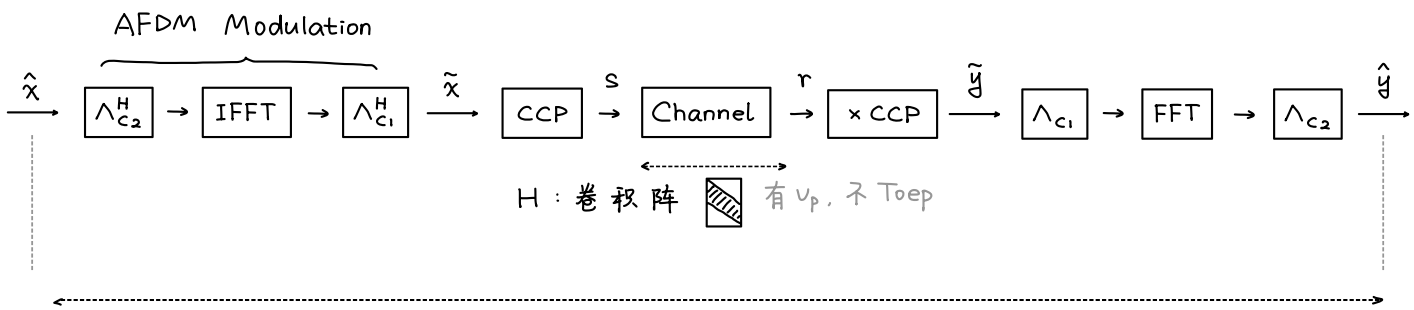
$$= \underbrace{F \tilde{H} F^H}_{\hat{H} = \mathrm{diag}(F h_{pad})} \hat{x} + \hat{n} = \mathrm{diag}(\hat{x}) F h_{pad} + \hat{n}$$



$$\hat{H} = \sum_{p=1}^{P} h_p \Omega^{\ell_p}$$

$$= \mathrm{diag}(F h_{pad})$$

AFDM 可同时处理 Double Dispersive ($\tau_p + \nu_p$)

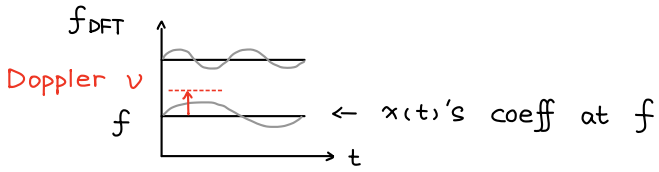


$\hat{H}$ 近似对角

$\hat{H} = \sum h_p \cdot \hat{H}_p$

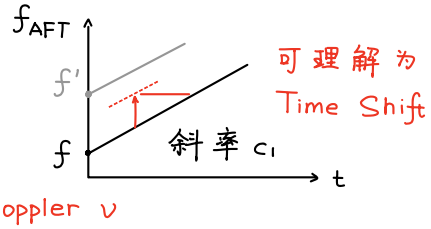
Fig 3 in AFDM 2021 Bemani

DFT basis $e^{j2\pi f t}$



AFT (Affine Fourier) / LCT (Linear Cononcial)

$$\hat{x}(f) = \langle \tilde{x}(t), e^{j2\pi[\underbrace{(c_1 t + f)}_{\text{Chirp 线性变频}} t + c_2]} \rangle$$

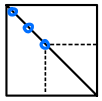


DAFT $\hat{x}_N = \underbrace{\Lambda_{c_2} F \Lambda_{c_1}}_{A} \tilde{x}_N$

$\Lambda_c = \mathrm{diag}(e^{j2\pi c n^2})$

AFDM

$$\tilde{y}_N = \sum_{p=1}^{P} h_p \cdot \underset{N\times N}{\Gamma_{CPP}^{p}} \cdot \underset{N\times N}{\overset{\mathrm{diag}(e^{j2\pi\nu_p n})}{\Delta^{\nu_p}}} \cdot \underset{N\times N}{\Pi^{\ell_p}} \cdot \tilde{x}_N + n = \tilde{H}\tilde{x} + n$$



ℓ_p

Π

$e^{-j2\pi c_1[N^2 - 2N(\ell_p - n)]}$

$$\hat{y} = A(\tilde{H}A^H\hat{x} + n)$$

$$= \sum_{p=1}^{P} h_p \cdot \underbrace{\Lambda_{c_2} F \Lambda_{c_1} \cdot \Gamma_{CPP}^{p} \Delta^{\nu_p} \Pi^{\ell_p} \cdot \Lambda_{c_1}^{H} F^H \Lambda_{c_2}^{H}}_{\hat{H}_p \ N\times N} \cdot \hat{x}_N + \hat{n}$$

在 OFDM 里本来应是 Ω^{ℓ_p}

$$= (h_P^T \otimes I_N)_{N\times NP} \cdot \begin{bmatrix} \hat{H}_1 \\ \vdots \\ \hat{H}_P \end{bmatrix}_{NP\times N} \cdot \hat{x}_N + \hat{n}$$

$$= [\hat{H}_1 x \cdots \hat{H}_P \hat{x}]_{N\times P} h_{P\times 1} + \hat{n}$$

估计 $\hat{x}$（Unknown h 但不关心）
估计 h（Known $\tilde{x}$）

$\begin{cases} \text{OFDM 均衡技术} \\ \text{AFDM } [] \end{cases}$

$$\min_h \ \| \hat{y} - \mathrm{diag}(\hat{x}) F h \|_2^2 \quad \text{s.t.} \ \| h \|_0 \le P$$

估计 h（Unknown $\tilde{x}$ 但不关心）

Blind Bistatic Radar（双盲雷达） 利用 $\sigma_{\tilde{x}}^2 = 1$

in Blind 2025 Ranasinghe

$$\min \ \| r_y - E \cdot \mathrm{vec}(H) \|^2$$
$$\text{s.t.} \ H \succeq 0$$
$$\mathrm{rank}(H) = 1,$$
$$\| \mathrm{vec}(H) \|_0 = P^2$$

Proposed：

$$\min \ \| r_y - E \cdot (\bar{h} \otimes h) \|^2$$
$$\text{s.t.} \ \| h \|_0 = P$$

ISAC（通感一体化）同时估计 $\hat{x}$ 和 h

$$\begin{cases} \text{OFDM} \rightarrow \text{Bilinear} \ ? \\ \text{AFDM} \ [\,] \end{cases}$$